

many memory CHOD(DtoH) $\Rightarrow$ many ^{too} hot $\Leftrightarrow$ device data copies / transfer
Not using pinned / memory or ~~asnc~~ data loaders
less than 10%

#4: black bar: CPU execute python script.
gray box: waiting or synchronization points

↓
for left-panel: CUDA APH: ~~cuda~~ $\left\{ \begin{array}{l} \text{cudaMemcpyAsync} \\ \text{cudaLaunchKernel} \\ \text{cudaStreamSynchronize} \end{array} \right.$

↓
Gaps between green kernel bars: GPU ~~not~~ idle gaps
 $\hookrightarrow$ Not enough overlapping of data transfer and compute

↓
too many blue / orange bars: frequent synchronization $\Rightarrow$ repeated
pthread_cond_wait below, CPU waiting for GPU sync or data copies

↓
gray / black boxes means to execute pytorch programs:

↓
Green bars short and overlap with compute $\Rightarrow \left\{ \begin{array}{l} \text{optimizer.step()} \\ \text{forward()} \\ \text{backward()} \end{array} \right.$
Almost $\rightarrow$ minimal idle time

↓
99.9% kernel: All GPU is to run kernel

0% Memory / memuse / DevH / HtoD: Almost zero time on data transfer

NCTX: Num of Context tokens a LLM can process in its attention

NVIX range: insert markers, ranges, and annotations directly ^{Window} into
CUDA code or python codes, to run a
Nvidia Tools extension profiler in Nsight / Nsight compute